

Tom Ciufu
Architecture Development 2

The commands I used to secure openEMR are:

1. Update System

```
sudo apt update && sudo apt upgrade -y
```

2. Enable Firewall (UFW)

```
sudo ufw allow OpenSSH
```

```
sudo ufw allow 80
```

```
sudo ufw allow 443
```

```
sudo ufw enable
```

```
sudo ufw status
```

3. Secure MySQL

```
sudo mysql_secure_installation
```

4. Secure File Permissions

```
sudo chown -R www-data:www-data /var/www/html/openemr
```

```
sudo find /var/www/html/openemr -type d -exec chmod 755 {} \;
```

```
sudo find /var/www/html/openemr -type f -exec chmod 644 {} \;
```

```
sudo chmod 640 /var/www/html/openemr/sites/default/sqlconf.php
```

5. Disable Directory Listing

Edit /etc/apache2/apache2.conf:

```
Options -Indexes +FollowSymLinks
```

```
sudo systemctl restart apache2
```

6. Enable HTTPS

```
sudo apt install certbot python3-certbot-apache -y
```

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```
sudo certbot --apache
```

7. Install Fail2Ban

```
sudo apt install fail2ban -y
```

```
sudo systemctl enable fail2ban
```

```
sudo systemctl start fail2ban
```

8. Disable Unnecessary Services

```
sudo systemctl list-units --type=service
```

```
sudo systemctl disable <service>
```

9. Secure SSH

Edit /etc/ssh/sshd_config:

```
PermitRootLogin no
```

```
PasswordAuthentication yes
```

```
sudo systemctl restart ssh
```

10. OpenEMR Internal Security

- Set strong admin password
- Enable ACL and audit logs
- Remove default users

Summary:

Security measures include firewall configuration, database hardening, file permission restrictions, intrusion prevention, HTTPS encryption, and service minimization.

I don't know what other attacks I would be susceptible to. To be honest, I don't even think I

did this assignment correctly. I had such a hard time with this one. Nothing worked right. I

spent 12 hours on this and I don't think I got it correct, but I will submit what I have